



Structure & Syllabus for Semester-V
Integrated M.Sc. (Computer Science) Programme
offered in
Department of Computer Science
Gujarat University
2025 – 2026

**As per NEP 2020 CURRICULUM AND CREDIT FRAMEWORK FOR
UNDERGRADUATE PROGRAMMES, UGC**

&

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of

Education Department, Govt. of Gujarat

STRUCTURE FOR SEMESTER - 5 COURSES IN GUJARAT UNIVERSITY FOR INTEGRATED M.Sc. (COMPUTER SCIENCE) AS PER NEP 2025 - 2026

COURSE:	INTEGRATED MASTER OF SCIENCE (COMPUTER SCIENCE)
MAJOR:	COMPUTER SCIENCE
MINOR: (Any One Track)	
	1. ARTIFICIAL INTELLIGENCE & MACHINE LEARNING (AIML) 2. INFORMATION SECURITY (IS) 3. WEB TECHNOLOGIES (WT)

SEMESTER - 5

MAJOR		
CODE	COURSE	CREDITS
DSC-C-CS-351T	MACHINE LEARNING	4
DSC-C-CS-352T	JAVA PROGRAMMING	4
DSC-C-CS-353P	JAVA PROGRAMMING – PRACTICALS	4
MINOR		
CODE	COURSE	CREDITS
DSC-M-CS-354T	DATA ANALYTICS	2
DSC-M-CS-354P	DATA ANALYTICS – PRACTICALS	2
DSC-M-CS-355T	ARTIFICIAL INTELLIGENCE	2
DSC-M-CS-355P	ARTIFICIAL INTELLIGENCE – PRACTICALS	2
	OR	
DSC-M-CS-354T	CRYPTOGRAPHY	2
DSC-M-CS-354P	CRYPTOGRAPHY – PRACTICALS	2
DSC-M-CS-355T	BLOCKCHAIN TECHNOLOGY	2
DSC-M-CS-355P	BLOCKCHAIN TECHNOLOGY – PRACTICALS	2
	OR	
DSC-M-CS-354T	WEB SECURITY	2
DSC-M-CS-354P	WEB SECURITY - PRACTICALS	2
DSC-M-CS-355T	ADVANCED WEB SCRIPTING	2
DSC-M-CS-355P	ADVANCED WEB SCRIPTING - PRACTICALS	2
INTER DISCIPLINARY / MULTI DISCIPLINARY		
CODE	COURSE	CREDITS
	NONE	
ABILITY ENHANCEMENT COURSE		
CODE	COURSE	CREDITS
	NONE	
SKILL ENHANCEMENT COURSE (ANY ONE COURSE)		
CODE	COURSE	CREDITS
SEC – 356	ADVANCED PYTHON PROGRAMMING – PRACTICALS	2
COMMON VALUE-ADDED COURSES		
CODE	COURSE	CREDITS
	NONE	

TOTAL CREDITS

22

5 Year Integrated MSc (Computer Science)
Exam Scheme for Semester – 5 (As per NEP)

Code	Course	Credits	Theory (Internal)	Theory (External)	Practical (Internal)	Practical (External)	Total
DSC-C-CS-351T	MACHINE LEARNING	4	50	50	***	***	100
DSC-C-CS-352T	JAVA PROGRAMMING	4	50	50	***	***	100
DSC-C-CS-353P	JAVA PROGRAMMING – PRACTICALS	4	***	***	50	50	100
DSC-M-CS-354T	DATA ANALYTICS	2	25	25	***	***	50
DSC-M-CS-354P	DATA ANALYTICS – PRACTICALS	2	***	***	25	25	50
DSC-M-CS-355T	ARTIFICIAL INTELLIGENCE	2	25	25	***	***	50
DSC-M-CS-355P	ARTIFICIAL INTELLIGENCE – PRACTICALS	2	***	***	25	25	50
DSC-M-CS-354T	CRYPTOGRAPHY	2	25	25	***	***	50
DSC-M-CS-354P	CRYPTOGRAPHY – PRACTICALS	2	***	***	25	25	50
DSC-M-CS-355T	BLOCKCHAIN TECHNOLOGY	2	25	25	***	***	50
DSC-M-CS-355P	BLOCKCHAIN TECHNOLOGY – PRACTICALS	2	***	***	25	25	50
DSC-M-CS-354T	WEB SECURITY	2	25	25	***	***	50
DSC-M-CS-354P	WEB SECURITY - PRACTICALS	2	***	***	25	25	50
DSC-M-CS-355T	ADVANCED WEB SCRIPTING	2	25	25	***	***	50
DSC-M-CS-355P	ADVANCED WEB SCRIPTING – PRACTICALS	2	***	***	25	25	50
SEC – 356	ADVANCED PYTHON – PRACTICALS	2	***	***	25	25	50
TOTAL		22	150	150	125	125	550

Semester - V

(Integrated M.Sc. (Computer Science) Programme)

MAJOR: COMPUTER SCIENCE

GUJARAT UNIVERSITY

INTEGRATED M.Sc. (COMPUTER SCIENCE)

PROGRAMME OUTCOMES

- **PO1:** To inculcate self-learning and life-long learning capabilities
- **PO2:** To become an employable and successful entrepreneur
- **PO3:** To create skilled human resources in the area of computer science that caters the need of software development and system analysts
- **PO4:** To create skilled human resources that can work as software developers
- **PO5:** To empower the students with efficient communication skills for project management
- **PO6:** To inculcate capabilities like modelling business problems and business logic into software systems
- **PO7:** To inculcate leadership and teamwork qualities
- **PO8:** To inculcate qualities that helps to develop sustainable solutions to interdisciplinary global problems through research and innovation capabilities
- **PO9:** To promote independent and collaborative work in the area of computer applications
- **PO10:** To provide fundamental knowledge of computing

PROGRAMME SPECIFIC OUTCOMES

- **PSO1:** To inculcate and apply latest technologies to solve business problems
- **PSO2:** To apply algorithmic approach and computer science techniques in designing computer-based solutions and applications
- **PSO3:** To apply technical knowledge for developing secure, smart and sustainable solutions
- **PSO4:** To apply the knowledge of software engineering in designing effective business solutions
- **PSO5:** To be able to create, maintain and troubleshoot the business solutions
- **PSO6:** To be able to enhance their skills through life-long learning through professional activities
- **PSO7:** To develop ability for effective communication with the team members and management
- **PSO8:** To develop ability to work in multidisciplinary area as a team member and to develop leadership qualities and managerial skills
- **PSO9:** To develop effective solutions using technologies that can scale and provide high performance solutions
- **PSO10:** To exhibit professional and personal ethics in the industry

Course Name: Machine Learning

Course Code: DSC-C-CS-351T

Credits: 4

Course Outcomes:

The aim of this course is to enable students to

- **CO1:** Develop a strong theoretical foundation for understanding of the state of art Machine Learning Algorithms
- **CO2:** Identify, formulate and solve machine learning problems that arise in practical applications
- **CO3:** Explore various tools and technologies that enable Machine Learning

Prerequisites:

Mathematical Foundations, Programming in C++/Python

Contents:

Unit No.	Particulars	Credits	Time
1.	Unit Title: Overview of Machine Learning Introduction to Machine learning from data, <i>Types of Machine Learning:</i> Supervised, Unsupervised, Reinforcement Learning, Concepts of Regression, Classification, Clustering	01	15 Hrs.
2.	Unit Title: Linear Regression & Logistic Regression Scatter diagram, Model representation for single variable, Single variable Cost Function, Least Square line fit, Normal Equations, Gradient Descent method for Linear Regression, Assumptions in Linear Regression, Properties of Regression line, Model Performance through R ² , Multivariable model representation, Multivariable cost function, Multiple Linear Regression, Overfitting, Underfitting, Bias and variance, Regularization, Issues of using Linear Regression in Classification, Sigmoid function, Odds of an event, Logit function, Decision Boundary, Maximum likelihood function, Linear Regression verses Logistic Regression, Pros and Cons of Logistic Regression, Cost function, Multi-classification, Confusion Matrix, <i>Statistical measures to measure Classification:</i> Recall, Precision, Accuracy	01	15 Hrs.
3.	Unit Title: Supervised Learning Classification problems, Decision Boundaries, K-nearest Neighbour methods, Linear Classifiers, Bayes' Rule and Naive Bayes Model, SVM - Introduction, Support Vectors & Margin, Optimization Objective, Dimensionality Reduction, PCA, Linear & Non-Linear SVM, Hard Margin & Soft Margin, Large Margin Classifiers, Kernels, Decision Tree, <i>Ensemble methods for Classification and Regression:</i> Bagging, <i>Boosting:</i> AdaBoost, Random Forest	01	15 Hrs.

Unit No.	Particulars	Credits	Time
4.	Unit: Unsupervised learning Cluster Analysis, Classification and Clustering, Definition of Clusters, Clustering Applications, Distance Measures, Proximity Measures for Discrete Variables, Proximity Measures for Mixed Variables, Partitional Clustering, Clustering Criteria, K-Means Algorithm, Fuzzy Clustering, Hierarchical Clustering, Agglomerative Hierarchical Clustering, Divisive Hierarchical Clustering, Cluster validity, External criteria, Internal criteria	01	15 Hrs.

References:

1. Mitchell T., "Machine Learning", McGraw Hill Publications
2. Grus J., "Data Science from Scratch", O'Reilly Publications
3. Alpaydin E., "Introduction to Machine Learning", MIT Press
4. Coelho R., "Building Machine Learning Systems with Python", Packt Publishing Ltd.
5. Marsland S., "MACHINE LEARNING: An Algorithmic Perspective", CRC Press
6. Wunsch D., Xu R., "Clustering", IEEE Press
7. Ethem Alpaydin, "Introduction to Machine Learning", The MIT Press

Accomplishments of the student after completing the course:

At the end of the course, students will be able to

- Develop an appreciation for what is involved in learning models from data
- Understand a wide variety of learning algorithms
Understand how to evaluate models generated from data
- Understand and develop application involving computer vision and Natural Language Processing
